

\CurrentAudience: Fac

PH113 / Assignment - Module V: Modern Physics

(Dated: April 9, 2024)

1. Outline the differences between spontaneous and stimulated emission.

Solution.

TABLE I. Difference between stimulated and spontaneous emission

	Stimulated Emission	Spontaneous emission
1	An atom in the excited state is induced to return to the ground state, thereby resulting in two photons of same frequency and energy is called Stimulated emission	The atom in the excited state returns to the ground state thereby emitting a photon, without any external inducement is called Spontaneous emission.
2	The emitted photons move in the same direction and is highly directional	The emitted photons move in all directions and are random
3	The radiation is highly intense, monochromatic and coherent	The radiation is less intense and is incoherent
4	The photons are in phase, there is a constant phase difference	The photons are not in phase (i.e.) there is no phase relationship between them.
5	The rate of transition is given by $\frac{dN_2}{dt} = B_{21}N_2$ (related to Einstein's coefficient B)	The rate of transition is given by $\frac{dN_2}{dt} = A_{21}N_2$ (related to Einstein's coefficient A)

2. Find the ratio of population of two states in a He-Ne laser that produces a light of wavelength 6328 Å at 27 °C.

Solution.

$$\begin{aligned}\frac{N_2}{N_1} &= \exp\left(\frac{-(E_2 - E_1)}{k_B T}\right) \\ &= \exp\left(-\frac{h\nu}{k_B T}\right) \\ &= \exp\left(-\frac{hc}{\lambda} \frac{1}{k_B T}\right)\end{aligned}$$

Plug in the values and solve.

3. A Ruby laser emits light of wavelength 7000 Å. Find the ratio of population of two states at 300 °C and at 500 °C.

Solution. Use the same expression as in the above question.

4. A certain Ruby laser emits 1.00 J pulses of light whose wavelength is 694 nm. What is the minimum number of Cr^{3+} ions in the ruby?

Solution. The minimum number of Cr^{3+} ions present in the ruby will be equal to the number of photons. The number of photons is given by the total energy of the pulse divided by the energy of the photons. Therefore, the number of photons, and equivalently the number of Cr^{3+} ions is equal to:

$$\frac{E_{\text{pulse}}}{hc/\lambda} = \frac{E_{\text{pulse}}\lambda}{hc} = 3.48 \times 10^{18}.$$

5. A ruby laser emits pulses of light with a wavelength of 694.3 nm. Calculate the frequency of the laser light and the energy of one photon emitted by the laser.

Solution. Just use $\nu = c/\lambda$ and $E = h\nu$, where the terms have usual meanings.

6. A He-Ne laser emits light with a wavelength of 632.8 nm. Determine the energy of one photon emitted by this laser and compare it with the energy of a photon emitted by a ruby laser (given in the previous question).

Solution. Same as above.

7. Given the binding energy per nucleon for a nucleus is 8.5 MeV, calculate the binding energy for a nucleus of mass number 56.

Solution. Just multiply the two numbers to obtain the binding energy of the nuclei.